

Sanjeeb Tiwary

Data Engineer | ML Developer | GenAI/RAG Specialist
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LinkedIn | GitHub | Portfolio

Professional Summary

Data and ML professional with **2.5+ years of experience** building scalable data pipelines, ML systems, GenAI/RAG applications, OCR workflows, APIs, and cloud-based automation solutions. Skilled in transforming raw data into analytics, searchable knowledge systems, and production-ready AI services.

Skills

Core: Python, SQL, PySpark, FastAPI, AWS, Docker, PostgreSQL, Qdrant, Scikit-learn, XGBoost, LangChain, RAG, OCR, ETL, MLflow

Data & ML: Data Cleaning, EDA, Feature Design, Classification, Forecasting, Anomaly Detection, Model Evaluation, Explainable AI

Tools: S3, Glue, Lambda, EC2, SageMaker, Apache Airflow, Git, CI/CD, Streamlit, REST APIs

Professional Experience

Technowell Enterprise Services Pvt. Ltd.

Jan 2024 – Present

ML & AI Specialist / Data Science Specialist

Hyderabad, India

- **Engineered** scalable data and ML pipelines across ingestion, transformation, validation, and serving, improving processing efficiency by **30%**.
- **Automated** ETL workloads using AWS Glue, Lambda, S3, PostgreSQL, and Spark, reducing manual reporting effort by **40%**.
- **Designed** anomaly detection and risk-scoring logic to flag outliers, prioritize alerts, and reduce manual review effort by **25%**.
- **Developed** real-time FastAPI inference services for ML outputs and document intelligence, refined API response handling by **35%**.
- **Implemented** OCR and document parsing workflows for scanned files, metadata extraction, and searchable retrieval across **1,000+ pages**.
- **Built** RAG-based GenAI applications with embeddings, vector search, prompt workflows, and citations, boosted document search accuracy by **30%**.
- **Optimized** model evaluation using precision, recall, F1-score, ROC-AUC, and threshold tuning, upgrading decision quality by **20%**.

Projects

- **Legal Bot LLM – RAG-Based Legal AI Assistant**
 - **Developed** a local GenAI legal assistant using FastAPI, Qdrant, PostgreSQL, Streamlit, Ollama, and RAG.
 - **Created** document ingestion for PDF/DOCX/TXT files with extraction, chunking, embedding, retrieval, and citation generation.
 - **Reduced** manual legal document review effort by **50%** through searchable knowledge retrieval and AI-based responses.
- **Fraud Risk Analytics & Anomaly Detection**
 - **Designed** workflows to detect suspicious behavior, high-risk signals, and abnormal patterns from structured datasets.
 - **Applied** classification, clustering, rule-based scoring, threshold tuning, and false-positive reduction techniques.
 - **Improved** alert prioritization by **25%** using anomaly scores, risk indicators, and business-rule logic.
- **OCR-Based Document Intelligence System**
 - **Implemented** OCR pipelines using Python, OpenCV, OCR models, and APIs for scanned documents and image-based text extraction.
 - **Enabled** searchable text extraction and metadata generation, reducing document processing time by **35%**.

Certifications & Publications

- **Earned** certifications in Machine Learning, SQL Analytics, and Generative AI from Udemy, DeepLearning.AI, and Coursera.
- **Published** research work in Springer, IEE, and ACM IC3 on ML, computer vision datasets, and prediction systems.

Education

Kalinga Institute of Industrial Technology (KIIT)

2024

B.Tech in Computer Science

CGPA: 7.9

KIIT Polytechnic / SCTE & VT

2021

Diploma in Electronics and Telecommunication

Percentage: 82%